ST3247 Midterm: Things to Know

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This document summarizes the basic definitions, formulas, techniques, and coding patterns that you should at the very minimum know for the ST3247 midterm examination.

1 Definitions to Know

1.1 Probability distributions

You should be comfortable with the following distributions: their densities, supports, and basic properties.

- **Uniform** on $[a, b]$: density $f(x) = \frac{1}{b-a}$ for $x \in [a, b]$. For $[0, 1]$: $\mathbb{E}[X] = 1/2$, $\mathrm{Var}(X) = 1/12$.
- **Gaussian (Normal)** $\mathcal{N}(\mu, \sigma^2)$: density $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)$. Light tails (exponential decay).
- **Exponential** with rate $\lambda > 0$: density $f(x) = \lambda e^{-\lambda x}$ for $x > 0$. Mean $= 1/\lambda$. CDF $= 1 - e^{-\lambda x}$.
- **Cauchy**: density $f(x) = \frac{1}{\pi(1+x^2)}$. Heavy tails (polynomial decay). The mean is *undefined*: the integral $\int x\, f(x)\, dx$ does not converge. Averaging n i.i.d. Cauchy random variables produces another Cauchy random variable.
- **Beta**$(2, 2)$: density $f(x) \propto x(1 - x)$ on $[0, 1]$.
- **Weibull** with shape $\alpha > 0$, scale $\lambda > 0$: CDF $F(x) = 1 - \exp\big(-(x/\lambda)^\alpha\big)$.
- **Student-t** with d degrees of freedom: density $p(x) \propto (1 + x^2/d)^{-(d+1)/2}$. For $d = 3$: $p(x) \propto (1 + x^2/3)^{-2}$, $\mathrm{Var}(X) = 3$. Heavy tails (polynomial decay).

1.2 Core concepts

- **PDF** (density): a function $f(x) \geq 0$ with $\int f(x)\, dx = 1$ such that $\mathbb{P}(a < X < b) = \int_a^b f(x)\, dx$.
- **CDF**: $F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f(t)\, dt$. Maps the real line to $[0, 1]$.

- **Inverse CDF**: $F^{-1}(u)$ solves $F(x) = u$. Central to the inversion method.
- **Normalizing constant**: $Z = \int p_u(x)\,dx$ so that $p(x) = p_u(x)/Z$ integrates to one.
- **Unnormalized density**: $p_u(x)$ where $p(x) = p_u(x)/Z$ and Z may be unknown.
- **Indicator function**: $\mathbf{1}_A(x) = 1$ if $x \in A$, and 0 otherwise.
- **i.i.d.**: independent and identically distributed random variables.
- **Importance ratio**: $r(x) = p(x)/q(x)$, the density ratio between target p and proposal q.
- **Unnormalized importance ratio**: $r_u(x) = p_u(x)/q_u(x)$.
- **Normalized importance weights**: $w_i = r_u(X_i)/\sum_{j=1}^{N} r_u(X_j)$. These are nonnegative and sum to one.
- **Effective Sample Size** (ESS): ESS $= 1/\sum_{i=1}^{N} w_i^2$, where w_i are normalized weights. Always satisfies $1 \leq \text{ESS} \leq N$.

2 Key Formulas to Know

2.1 Sample mean

Given $X_1, \ldots, X_N$ i.i.d. with mean μ and variance σ^2, the sample mean is

$$\bar{X}_N = \frac{1}{N}\sum_{i=1}^{N} X_i, \qquad \mathbb{E}[\bar{X}_N] = \mu, \qquad \mathrm{Var}(\bar{X}_N) = \frac{\sigma^2}{N}.$$

The variance decreases as $1/N$: more samples give a more precise estimate.

2.2 Law of Large Numbers (LLN)

$\bar{X}_N \to \mu$ as $N \to \infty$ (convergence in probability). The sample mean converges to the true mean.

2.3 Central Limit Theorem (CLT)

For large N, the sample mean is approximately Gaussian:

$$\bar{X}_N \;\approx\; \mathcal{N}\left(\mu,\; \frac{\sigma^2}{N}\right).$$

This is the foundation for constructing confidence intervals.

2.4 Monte Carlo integral approximation

Any integral over $[a, b]$ (or $[a, b]^d$) can be rewritten as an expectation:

$$\int_a^b \varphi(x)\, dx \;=\; (b-a) \cdot \mathbb{E}[\varphi(U)], \qquad U \sim \text{Uniform}(a, b).$$

The Monte Carlo estimate is $(b-a) \cdot \frac{1}{N} \sum_{i=1}^{N} \varphi(U_i)$. For integrals over $[a, b]^d$, multiply by the volume $(b-a)^d$.

3 Techniques to Master

3.1 Basic Monte Carlo

The core idea: express a quantity of interest as an expectation, then approximate it by averaging random samples.

Integrals. Rewrite $\int_a^b \varphi(x)\, dx$ as $(b-a) \cdot \mathbb{E}[\varphi(U)]$ with $U \sim \text{Uniform}(a, b)$, then average N evaluations. For integrals over $[a, b]^d$, multiply by volume $(b-a)^d$.

Probabilities. Since $\mathbb{P}(\text{event}) = \mathbb{E}[\mathbf{1}_{\text{event}}]$, any probability can be estimated by the proportion of samples that satisfy the event.

Expectations. $\mathbb{E}[g(X)] \approx \frac{1}{N} \sum_{i=1}^{N} g(X_i)$ where $X_i \sim p$.

Volume estimation. To estimate the volume of $V \subset [a, b]^d$: sample uniformly on $[a, b]^d$, count the proportion of points falling in V, and multiply by $(b-a)^d$. For example, $\text{Vol}\{0 < x < y < z < 2\} = 8 \cdot \mathbb{P}(U_1 < U_2 < U_3)$ where $U_i \sim \text{Uniform}(0, 2)$.

Conditional expectations. To estimate $\mathbb{E}[g(X) \mid \text{event}]$: simulate many samples, discard those not satisfying the event, and average g over the remaining ones. For example, $\mathbb{E}[XY \mid X + Y > 1]$ with $X, Y \sim \mathcal{N}(0, 1)$: generate pairs (X_i, Y_i), keep those with $X_i + Y_i > 1$, and average $X_i Y_i$ over the kept samples.

Recognizing distributions in integrands. A powerful trick: factor out a known density from the integrand, then sample from that density. Three important cases:

$$\int \varphi(x)\, e^{-x^2/2}\, dx = \sqrt{2\pi} \cdot \mathbb{E}[\varphi(X)], \qquad X \sim \mathcal{N}(0,1),$$

$$\int \frac{\varphi(x)}{1+x^2}\, dx = \pi \cdot \mathbb{E}[\varphi(X)], \qquad X \sim \text{Cauchy},$$

$$\int \varphi(x)\, e^{-x}\, dx = \mathbb{E}[\varphi(X)], \qquad X \sim \text{Exp}(1).$$

3.2 Inversion method

The inversion method transforms a uniform random variable into a sample from any distribution whose CDF can be inverted analytically. The procedure is:

1. Compute the CDF: $F(x) = \int f(t)\, dt$.
2. Invert it: solve $F(x) = u$ for x to obtain F^{-1}.
3. Sample: set $X = F^{-1}(U)$ where $U \sim \text{Uniform}(0,1)$.

Target density	**CDF** $F(x)$	**Inverse CDF** $F^{-1}(u)$
$f(x) \propto x^4$ on $[0,1]$	x^5	$u^{1/5}$
$f(x) \propto x^3$ on $[0,2]$	$x^4/16$	$2\,u^{1/4}$
Cauchy	$\frac{1}{\pi}\arctan(x) + \frac{1}{2}$	$\tan(\pi(u - \frac{1}{2}))$
Exponential (rate λ)	$1 - e^{-\lambda x}$	$-\log(1-u)/\lambda$
Weibull (shape α, scale λ)	$1 - e^{-(x/\lambda)^\alpha}$	$\lambda\big(-\log(1-u)\big)^{1/\alpha}$

The method is limited to distributions whose CDF can be inverted in closed form. For example, the Gaussian CDF has no closed-form inverse, so the inversion method cannot be applied directly.

3.3 Rejection sampling

Rejection sampling allows sampling from a density $f(x) \propto \tilde{f}(x)$ by using a simpler proposal distribution. It works even when the normalizing constant of f is unknown.

1. Find a proposal density f_W and a constant $c > 0$ such that $\tilde{f}(x) \le c \cdot f_W(x)$ for all x.

2. Generate $W \sim f_W$ and $U \sim \text{Uniform}(0,1)$.
3. Accept $X = W$ if $U \leq \tilde{f}(W)\big/(c \cdot f_W(W))$; otherwise reject and go back to step 2.

The expected number of proposals before acceptance is $\alpha = \int c \cdot f_W(x)\,dx \geq 1$. A tighter envelope gives a higher acceptance rate.

Example: to sample a standard Gaussian using a Cauchy proposal, find c by maximizing the ratio $\tilde{f}_{\text{Gauss}}(x)/f_{\text{Cauchy}}(x)$.

3.4 Importance sampling (IS)

Importance sampling is used to estimate expectations under a target distribution p by sampling from a different proposal distribution q. It is based on the identity

$$\mu = \int \varphi(x)\,p(x)\,dx = \int \varphi(x)\,\frac{p(x)}{q(x)}\,q(x)\,dx = \mathbb{E}_q\big[\varphi(X)\,r(X)\big],$$

where $r(x) = p(x)/q(x)$ is the importance ratio and $X \sim q$.

Standard IS estimator.

1. Choose a proposal q with $q(x) > 0$ wherever $p(x) > 0$.
2. Sample $X_1, \ldots, X_N \sim q$.
3. Compute $\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} \varphi(X_i) \cdot r(X_i)$.

This estimator is **unbiased**: $\mathbb{E}[\hat{\mu}] = \mu$.

Self-Normalized IS (SNIS). When $p(x) = p_u(x)/Z_p$ with Z_p unknown, the standard IS ratio $r(x)$ cannot be evaluated. SNIS bypasses this by normalizing the weights:

1. Sample $X_1, \ldots, X_N \sim q$.
2. Compute unnormalized ratios $r_u(X_i) = p_u(X_i)/q_u(X_i)$.
3. Normalize: $w_i = r_u(X_i)\big/\sum_{j=1}^{N} r_u(X_j)$.
4. Estimate: $\hat{\mu} = \sum_{i=1}^{N} w_i \cdot \varphi(X_i)$.

SNIS is **biased** but **consistent**: the bias vanishes as $N \to \infty$ and is typically negligible compared to the standard deviation.

Ratio of normalizing constants. A useful identity that underlies SNIS: for $p(x) = p_u(x)/Z_p$ and $q(x) = q_u(x)/Z_q$,

$$\frac{Z_p}{Z_q} \;=\; \mathbb{E}_q\left[\frac{p_u(X)}{q_u(X)}\right].$$

Optimal proposal. The proposal that minimizes the variance of the IS estimator is

$$q^*(x) \;\propto\; |\varphi(x)|\, p(x).$$

When $\varphi(x) \geq 0$, this choice gives *zero* variance. It is not practical (computing the normalizing constant of q^* requires the very integral we want to estimate), but it provides a useful guideline: *aim for a proposal whose shape roughly matches* $|\varphi(x)|\, p(x)$.

Choosing a good proposal. Three rules of thumb:

- The proposal should roughly match the shape of $|\varphi(x)|\, p(x)$.
- **Never use a proposal with lighter tails than the target.** A Gaussian proposal for a Student-t target is dangerous: the importance ratio $p(x)/q(x)$ grows without bound in the tails, producing infinite or near-infinite variance.
- Prefer heavy-tailed proposals (Cauchy, Student-t) as a safe default.

Effective Sample Size (ESS). The ESS measures the quality of the importance weights:

$$\text{ESS} = \frac{1}{\sum_{i=1}^{N} w_i^2}, \qquad 1 \leq \text{ESS} \leq N.$$

ESS $\approx N$ means the weights are roughly uniform (good); ESS ≈ 1 means a single weight dominates (bad, essentially a single-sample estimate).

Rare event estimation. Estimating $\mathbb{P}(X \in A)$ when A is a rare event is a classic application of IS. Naive MC wastes almost all samples (most $\mathbf{1}_{X \in A} = 0$). IS fixes this by choosing a proposal q concentrated on A and reweighting with $p(X_i)/q(X_i)$.

Example: to estimate $\mathbb{P}(X^2 + Y^2 > 40)$ with $X, Y \sim \mathcal{N}(0, 1)$, use the wider proposal $X_{\text{IS}}, Y_{\text{IS}} \sim \mathcal{N}(0, 5^2)$.

4 Python/NumPy Coding Skills

4.1 Random number generation

```
np.random.uniform(a, b, N)                # Uniform on [a, b]
np.random.normal(mu, sigma, N)             # Gaussian N(mu, sigma^2)
np.random.standard_cauchy(N)               # Standard Cauchy
np.random.exponential(scale=mu, size=N)    # Exponential with mean mu
np.random.randint(low=a, high=b)           # Integer in {a, ..., b-1}
```

4.2 Boolean masking for conditional estimation

```
X = np.random.normal(0, 1, N)
Y = np.random.normal(0, 1, N)
mask = (X + Y) > 1
conditional_mean = np.mean(X[mask] * Y[mask])
```

4.3 SNIS weight computation pattern

```
# Target: f(x) proportional to log(1+x^2) on [0,3]
# Proposal: Uniform(0,3)
x = np.random.uniform(0, 3, N)
w_unnorm = np.log(1 + x**2)           # unnormalized weights
w = w_unnorm / np.sum(w_unnorm)        # normalized weights
E_X = np.sum(x * w)                    # SNIS estimate of E[X]
E_X2 = np.sum(x**2 * w)               # SNIS estimate of E[X^2]
std_X = np.sqrt(E_X2 - E_X**2)        # standard deviation
```